SC18IS606

I²C-Bus to SPI Bridge

Rev. 1.2 — 13 October 2026 Product data sheet

1 General description

SC18IS606 is designed to serve as an interface between a standard Inter-Integrated Circuit bus (I²C-bus) of a microcontroller and a Serial Peripheral Interface (SPI) bus. This allows the microcontroller to communicate directly with SPI devices through its I²C-bus. SC18IS606 operates as an I²C-bus target-transmitter or target-receiver and an SPI controller. SC18IS606 controls all the SPI bus-specific sequences, protocol, and timing. SC18IS606 has its own internal oscillator, and it supports three SPI chip select outputs that can be configured as General-Purpose Input/Output (GPIO) when not used as SPI Chip Select (CS).

SC18IS606 is a functional replacement for SC18IS602B except for:

- New pinout
- Three instead of four chip selects; no quasi-bidirectional mode
- Lower operating supply voltage level (1.71 V vs 2.4 V)
- Data buffer increased to 1024 bytes vs 200 bytes
- Temperature range of -40 °C to +105 °C vs -40 °C to +85 °C
- The device is rotated 180 degrees in the tape pocket; pin 1 is now in Quadrant 1

2 Features and benefits

- I²C-bus target interface operating up to 400 kHz
- SPI controller operating up to 1.8 Mbit/s
- 1024-byte data buffer
- Up to three targets select outputs
- Up to three programmable I/O pins
- Operating supply voltage: 1.71 V to 3.6 V
- Low power mode
- Internal oscillator
- Active LOW interrupt output
- Electrostatic Discharge (ESD) protection exceeds 2000 V Human Body Model (HBM) per JESD22-A114
- Latch-up testing is done to JEDEC Standard JESD78 that exceeds 100 mA
- Available in 16-pin TSSOP

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3 Applications

- Converting I²C-bus to SPI
- Adding additional SPI bus controllers to an existing system

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4 Ordering information

Table 1 describes the ordering information of SC18IS606.

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Version
SC18IS606PW	18IS606	TSSOP16	Plastic thin shrink small outline package; 16 leads; body width 4.4 mm	SOT403-1

4.1 Ordering options

Table 2 describes the ordering options of SC18IS606.

Table 2. Ordering options

Type number	Orderable part number	Package	Packing method	Minimum order quantity	Temperature
SC18IS606PW	SC18IS606PWJ	TSSOP16	REEL 13" Q1 *STANDARD MARK SMD [1]	2500	T_{amb} = -40 °C to +105 °C

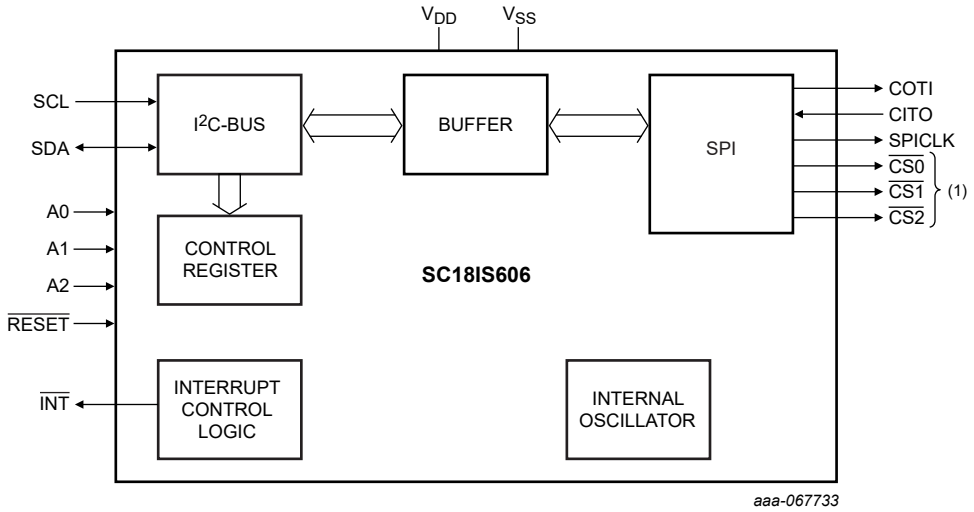
[1] Find packing information at www.nxp.com/docs/en/packing/SOT403-1_118.pdf.

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5 Block diagram

Figure 1 shows the block diagram of SC18IS606.



1. Unused chip select outputs may be used for GPIO.

Figure 1. Block diagram of SC18IS606

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6 Pinning information

This section provides the pin configuration and pin description of the SC18IS606.

6.1 Pinning

Figure 2 shows the pin configuration of SC18IS606.

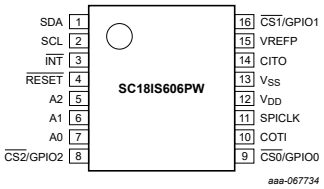


Figure 2. Pin configuration for TSSOP16

6.2 Pin description

Table 3 provides detailed description of various pins on SC18IS606.

Table 3. Pin description

Symbol	Pin	Type	Description
$\overline{CS0}$/GPIO0	9	I/O	SPI chip select output 0 (active LOW) or GPIO 0
$\overline{CS1}$/GPIO1	16	I/O	SPI chip select output 1 (active LOW) or GPIO 1
$\overline{RESET}$	4	I	Reset input (active LOW)
V_{SS}	13	-	Ground supply
CITO	14	I	Controller In, target Out
COTI	10	O	Controller Out, target In
SDA	1	I/O	I^2C-bus data
SCL	2	I	I^2C-bus clock
$\overline{INT}$	3	O	Interrupt output (active LOW). This pin is an open-drain pin, which must be pulled HIGH with a resistor and must not be held LOW at power on or reset
$\overline{CS2}$/GPIO2	8	I/O	SPI chip select output 2 (active LOW) or GPIO 2
SPICLK	11	O	SPI clock
V_{DD}	12	-	Supply voltage
VREFP	15	-	Must connect to V_{DD}
A0	7	I	Address input 0
A1	6	I	Address input 1
A2	5	I	Address input 2

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7 Functional description

SC18IS606 acts as a bridge between an I²C-bus and an SPI interface. It allows an I²C-bus controller device to communicate with any SPI-enabled device.

7.1 I²C-bus interface

The I²C-bus uses two wires (SDA and SCL) to transfer information between devices connected to the bus, and it has the following features:

- Bidirectional data transfer between controllers and targets
- Multi-controller bus (no central controller)
- Arbitration between simultaneously transmitting controllers without corruption of serial data on the bus
- Serial clock synchronization allows devices with different bit rates to communicate via one serial bus
- Serial clock synchronization can be used as a handshake mechanism to suspend and resume serial transfer
- The I²C-bus can be used for test and diagnostic purposes

A typical I²C-bus configuration is shown in Figure 3 (refer to NXP Semiconductors UM10204, I²C-bus specification and user manual).

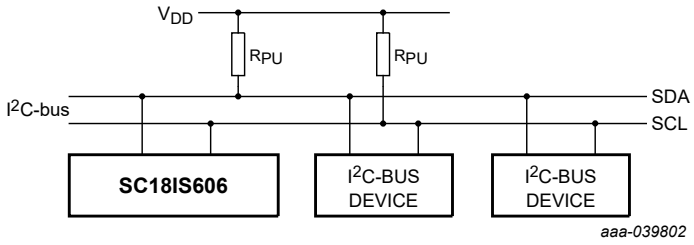


Figure 3. I²C-bus configuration

SC18IS606 device provides a byte-oriented I²C-bus interface that supports data transfers up to 400 kHz. When the I²C-bus controller is reading data from SC18IS606, the device is a target-transmitter. SC18IS606 is a target-receiver when the I²C-bus controller is sending data. At no time does SC18IS606 act as an I²C-bus controller, however, it can hold the SCL line LOW between bytes to complete its internal processes (for example, clock stretching).

7.1.1 Addressing

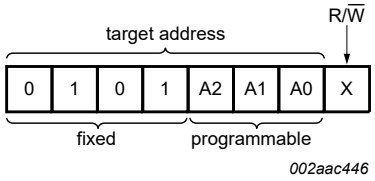


Figure 4. Target address

The first seven bits of the first byte sent after a START condition defines the target address of the device being accessed on the bus. The eighth bit determines the direction of the message. A ‘0’ in the least significant position of the first byte means that the controller writes information to a selected target. A ‘1’ in this position means that the controller reads information from the target. When an address is sent, each device in a system compares the first seven bits after the START condition with its address. If they match, the device considers itself addressed by the controller as a target-receiver or target-transmitter, depending on the R/$\overline{W}$ bit.

A target address of SC18IS606 consists of a fixed and a programmable part. The programmable part of the target address enables the maximum number of such devices to be connected to the I²C-bus. Since SC18IS606

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has three programmable address bits (defined by the A2, A1, and A0 pins), it is possible to have eight of these devices on the same bus.

The state of the A2, A1, and A0 pins are latched at reset. Changes made after reset do not alter the address.

When SC18IS606 is busy after the address byte is transmitted, it will not acknowledge its address.

7.1.2 Write to a data buffer

All communications to or from SC18IS606 occur through the data buffer. The data buffer is 1024 bytes deep. A message begins with SC18IS606 address, followed by the Function ID. Depending upon the Function ID, zero to 1024 data bytes can follow.

SC18IS606 places the received data into a buffer and continues loading the buffer until a STOP condition is received. After the STOP condition is detected, further communications are not acknowledged until the function designated by the Function ID has been completed.

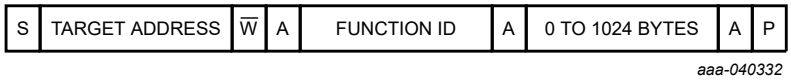


Figure 5. Write to a data buffer

7.1.3 SPI read and write - Function ID 01h to 0x7h

Data in the buffer is sent to the SPI port if the Function ID is 01h to 0x7h. The Function ID contains the Chip Select (CS) to be used for the transmission on the SPI port. There are three target selects that can be used, with each CS being selected by one of the bits in the Function ID. There is no restriction on the number or combination of CSs that can be enabled for an SPI message. If more than one $\overline{CSn}$ pin is enabled at one time, the user must be aware of possible contention on the data outputs of the SPI target devices.

Table 4. Function ID 01h to 0x7h

7	6	5	4	3	2	1	0
0	0	0	0	0	CS2	CS1	CS0

The data on the SPI port contains the same information as the I²C-bus data, but without the target address and Function ID. For example, if the message shown in Figure 6 is transmitted on the I²C-bus, the SPI bus sends the message shown in Figure 7.

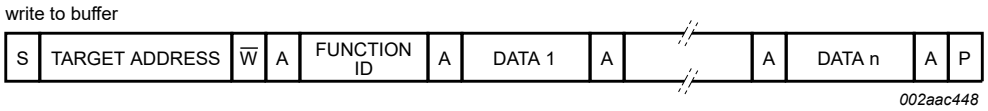


Figure 6. I²C-bus message

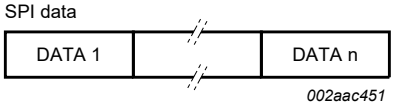


Figure 7. SPI message

SC18IS606 counts the number of data bytes sent to the I²C-bus port and automatically sends this same number of bytes to the SPI bus. As the data is transmitted from the COTI pin, it is also read from the CITO pin and saved in the data buffer. Therefore, the old data in the buffer is overwritten. The data in the buffer can then be read back.

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If the data from the SPI bus must be returned to the I²C-bus controller, the process must be completed by reading the data buffer. Section 8 gives an example of an SPI read.

7.1.4 Read from buffer

A read from the data buffer requires no Function ID. The target address with the R/$\overline{W}$ bit set to a '1' causes SC18IS606 to send the buffer contents to the I²C-bus controller. The buffer contents are not modified during the read process.

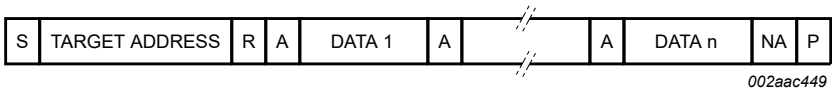


Figure 8. Read from buffer

A typical write and read from an SPI Electrically Erasable Programmable Read-Only Memory (EEPROM) is shown in Section 8.

After a read, the interrupt line is cleared.

7.1.5 Configure SPI Interface - Function ID F0h

The SPI hardware operating mode, data direction, and frequency can be changed by sending a **Configure SPI Interface** command to the I²C-bus.

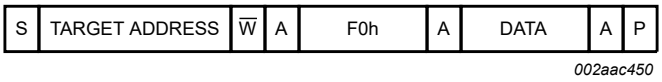


Figure 9. Configure SPI Interface

After SC18IS606 address is transmitted on the bus, the Configure SPI Interface Function ID (F0h) is sent followed by a byte, which defines the SPI communications.

The Clock Phase bit (CPHA) allows the user to set the edges for sampling and changing data. The Clock Polarity bit (CPOL) allows the user to set the clock polarity. Figure 18 and Figure 19 show the different settings of Clock Phase bit.

Table 5. Configure SPI Interface (F0h) bit allocation

Bit	7	6	5	4	3	2	1	0
Symbol	X	X	ORDER	X	MODE1	MODE0	F1	F0
Reset	X	X	0	X	0	0	0	0

Table 6. Configure SPI Interface (F0h) bit description

Bit	Symbol	Description
7:6	-	Reserved
5	ORDER	When logic 0, the MSB of the data word is transmitted first. If logic 1, the Least Significant Bit (LSB) of the data word is transmitted first.
4	-	Reserved
3:2	MODE1:MODE0	Mode selection 00 - SPICLK LOW when idle; data clocked in on leading edge (CPOL = 0, CPHA = 0) 01 - SPICLK LOW when idle; data clocked in on trailing edge (CPOL = 0, CPHA = 1)

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Table 6. Configure SPI Interface (F0h) bit description...*continued*

Bit	Symbol	Description
		10 - SPICLK HIGH when idle; data clocked in on trailing edge (CPOL = 1, CPHA = 0) 11 - SPICLK HIGH when idle; data clocked in on leading edge (CPOL = 1, CPHA = 1)
1:0	F1:F0	SPI clock rate 00 - 1875 kHz 01 - 455 kHz 10 - 115 kHz 11 - 58 kHz

7.1.6 Clear Interrupt - Function ID F1h

SC18IS606 generates an interrupt after completing any SPI transmission. This interrupt can be cleared ($\overline{INT}$ pin HIGH) by sending a **Clear Interrupt** command. It is not necessary to clear the interrupt; when polling the device, this function can be ignored.

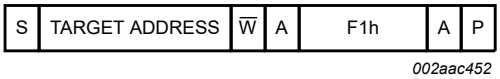


Figure 10. Clear Interrupt

7.1.7 Idle mode - Function ID F2h

A low-power mode can be entered by sending the **Idle Mode** command.

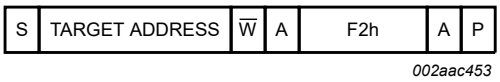


Figure 11. Idle mode

The Idle mode is exited when its I²C-bus address is detected.

7.1.8 GPIO Write - Function ID F4h

The state of the pins defined as GPIO can be changed using the Port Write function.

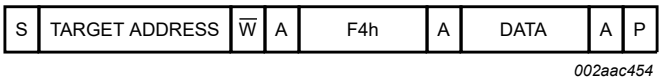


Figure 12. GPIO Write

The data byte following the F4h command determines the state of CS2, CS1, and CS0, if they are configured as GPIO. If these pins are used as SPI target selects or if they are GPIO, the Port Enable function defines.

Table 7. GPIO Write (F4h) bit allocation

Bit	7	6	5	4	3	2	1	0
Symbol	X	X	X	X	X	CS2	CS1	CS0
Reset	X	X	X	X	X	0	0	0

7.1.9 GPIO Read - Function ID F5h

The state of the pins defined as GPIO can be read into SC18IS606 data buffer using the GPIO Read function.

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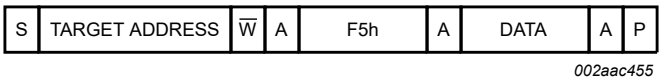


Figure 13. GPIO Read

Note: *This function does not return the value of the GPIO. To receive the GPIO contents, a one-byte Read Buffer command would be required. The value of the Read Buffer command returns the following byte.*

Table 8. GPIO Read (F5h) bit allocation

7	6	5	4	3	2	1	0
X	X	X	X	X	CS2	CS1	CS0

Data for pins not defined as GPIO are undefined.

A GPIO Read is always performed to update the GPIO data in the buffer. The buffer is undefined after the GPIO data is read back from the buffer. Therefore, reading data from the GPIO always requires a two-message sequence (GPIO Read, followed by Read Buffer).

7.1.10 GPIO Enable - Function ID F6h

At reset, the chip select pins ($\overline{CS0}$, $\overline{CS1}$, and $\overline{CS2}$) are configured to be used as chip select outputs. If these pins are not required for the SPI functions, they can be used as GPIO after they are enabled as GPIO. Any combination of pins can be configured to function as GPIO or chip selects.

After the GPIO Enable function is sent, the ports defined as GPIO are configured as input-only.

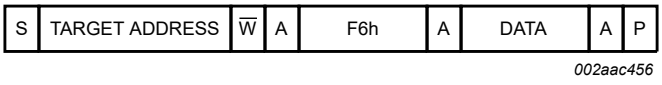


Figure 14. GPIO Enable

The data byte following the F6h command byte determines which pins can be used as GPIO. A logic 1 enables the pin as a GPIO, while a logic 0 disables GPIO control.

Table 9. GPIO Enable (F6h) bit allocation

7	6	5	4	3	2	1	0
X	X	X	X	X	CS2	CS1	CS0

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7.1.11 GPIO Configuration - Function ID F7h

The pins defined as GPIO can be configured by software to one of three types on a pin-by-pin basis. These are: push-pull, open-drain, and input-only.

Two bits select the output type for each port pin.

Table 10. GPIO Configuration (F7h) bit allocation

7	6	5	4	3	2	1	0
X	X	CS2.1	CS2.0	CS1.1	CS1.0	CS0.1	CS0.0

Table 11. GPIO Configuration (F7h) bit description

Bit	Symbol	Description
7	X	
6	X	
5	CS2.1	CS2[1:0] = 00: input-only (high-impedance) CS2[1:0] = 01: push-pull CS2[1:0] = 10: input-only (high-impedance) CS2[1:0] = 11: open-drain
4	CS2.0	
3	CS1.1	CS1[1:0] = 00: input-only (high-impedance) CS1[1:0] = 01: push-pull CS1[1:0] = 10: input-only (high-impedance) CS1[1:0] = 11: open-drain
2	CS1.0	
1	CS0.1	CS0[1:0] = 00: input-only (high-impedance) CS0[1:0] = 01: push-pull CS0[1:0] = 10: input-only (high-impedance) CS0[1:0] = 11: open-drain
0	CS0.0	

The $\overline{CSn}$ pins defined as GPIO, for example CS0.0 and CS0.1, can be configured by software to one of three types. These are: push-pull, open-drain, and input-only. Two configuration bits in the GPIO Configuration register for each pin select the type for each pin. A pin has a Schmitt-triggered input that also has a glitch suppression circuit.

7.1.11.1 Open-drain output configuration

The open-drain output configuration turns off all pull-ups and only drives the pulldown transistor of the pin when the port latch contains a logic 0. To be used as a logic output, a pin configured in this manner must have an external pullup, typically a resistor tied to V_{DD}.

The open-drain pin configuration is shown in Figure 15.

An open-drain pin has a Schmitt-triggered input that also has a glitch suppression circuit.

The device uses a pseudo open-drain mode. The pin cannot be pulled up above V_{DD}. The pins are not 5 V tolerant when V_{DD} is grounded.

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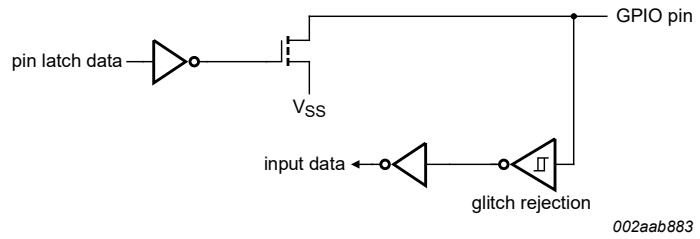


Figure 15. Open-drain output configuration

7.1.11.2 Input-only configuration

The input-only pin configuration is shown in Figure 16. It is a Schmitt-triggered input that also has a glitch suppression circuit.

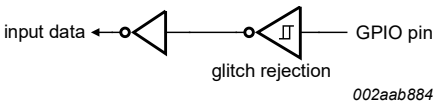


Figure 16. Input-only configuration

7.1.11.3 Push-pull output configuration

The push-pull output configuration has the same pulldown structure as the open-drain but provides a continuous strong pullup when the port latch contains a logic 1. The push-pull mode can be used when source current is needed from a pin output.

The push-pull pin configuration is shown in Figure 17.

A push-pull pin has a Schmitt-triggered input that also has a glitch suppression circuit.

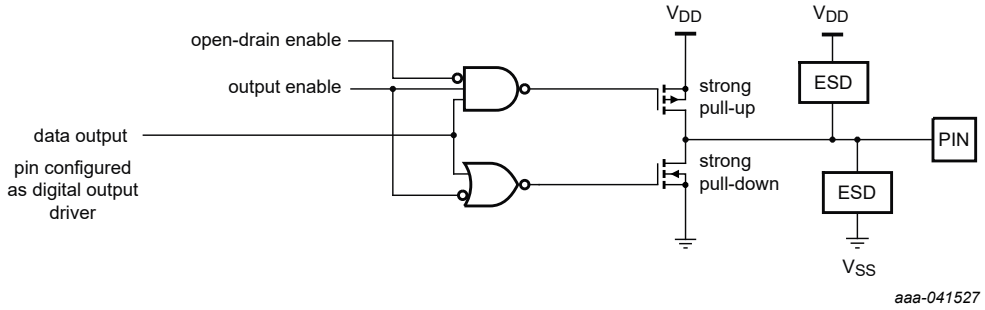


Figure 17. Push-pull output configuration

7.1.12 Read Version: Function ID FEh

This function commands the SC18IS606 to place version information into the read buffer. Once this command is issued, a null terminated string is placed in the read buffer. The string contains the part number and a version string “SC18IS606 1.0.0”.

The total length is 16 bytes including a 00h null terminator. Values in the Read buffer past the null terminator remain from a previous operation. The example above yields these hex values in the first 16 locations in the read buffer: 0x53 0x43 0x31 0x38 0x49 0x53 0x36 0x30 0x36 0x20 0x31 0x2e 0x30 0x2e 0x30 0x00

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7.2 SPI interface

The SPI interface can support Mode 0 through Mode 3 of the SPI specification and can operate up to 1.8 Mbit/s. The SPI interface uses at least four pins: SPICLK, COTI, CITO, and $\overline{CSn}$.

$\overline{CSn}$ are the chip select pins. In a typical configuration, an SPI controller selects one SPI device as the current target.

There are actually three $\overline{CSn}$ pins ($\overline{CS0}$, $\overline{CS1}$, and $\overline{CS2}$) to allow SC18IS606 to communicate with multiple SPI devices.

SC18IS606 generates the SPICLK (SPI clock) signal to send and receive data. The SCLK, COTI, and CITO are typically tied together between two or more SPI devices. Data flows from SC18IS606 (controller) to target on the COTI pin (pin 10) and the data flows from target to SC18IS606 (controller) on the CITO pin (pin 14).

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8 I²C-bus to SPI communications example

The following example describes a typical sequence of events required to read the contents of an SPI-based EEPROM. This example assumes that SC18IS606 is configured to respond to address 50h. A START condition is shown as ‘ST’, while a STOP condition is ‘SP’. The data is presented in hexadecimal format.

This sequence is not included in the testing, as there is no EEPROM on the bridge board, but it must work as written. A script for reading SPI Flash was tested, however.

1. The first message is used to configure the SPI port for mode and frequency.
   `ST,50,F0,02,SP SPI frequency 115 kHz using Mode 0`
2. An SPI EEPROM first requires that a Write Enable command be sent before data can be written.
   `ST,50,04,06,SP EEPROM write enable using CS2, assuming the Write Enable is 06h`
3. Clear the interrupt. This is not required if using a polling method rather than interrupts.
   `ST,50,F1,SP Clear interrupt`
4. Write the 8 data bytes. The first byte (Function ID) tells SC18IS606, which chip select output to use. This example uses CS2 (shown as 04h). The first byte sent to the EEPROM is normally 02h for the EEPROM write command. The next one or two bytes represent the subaddress in the EEPROM. In this example, a two-byte subaddress is used. Bytes 00 and 30 would cause the EEPROM to write to subaddress 0030h. The next eight bytes are the eight data bytes that are written to subaddresses 0030h through 0037h.
   `ST,50,04,02,00,30,01,02,03,04,05,06,07,08,SP Write 8 bytes using CS2`
5. When an interrupt occurs, do a Clear Interrupt or wait until SC18IS606 responds to its I²C-bus address.
   `ST,50,F1,SP Clear interrupt`
6. Read the eight bytes from the EEPROM. Note that a command is written, even though a read from the SPI port is performed. The Function ID is again 04h, indicating that CS2 is used. The EEPROM requires a 03h for a read, followed by the subaddress to be read. The same data previously written is read back, so the subaddress is 0030h. To read back eight bytes, eight bytes of FFh are sent to tell SC18IS606 to send eight more bytes on COTI. While it is sending these eight data bytes, it is also reading the CITO pin and saving the data in the buffer.
   `ST,50,04,03,00,30,FF,FF,FF,FF,FF,FF,FF,FF,SP Read 8 bytes using CS2`
7. The interrupt can be cleared, if needed.
   `ST,50,F1,SP Clear interrupt`
8. Read back the data buffer. Note that it is necessary to read back 11 data bytes since the first three bytes sent on the SPI port are the read code (03h) and the two subaddress bytes.
   `ST,50,00,00,00,01,02,03,04,05,06,07,08,SP Read the data buffer`
   It can be seen that on the I²C-bus the first four bytes do not contain the data from the SPI bus. The first byte is SC18IS606 address, followed by three dummy data bytes. These dummy data bytes correspond to the three bytes sent to the EEPROM before it actually places data on the bus (command 03h, subaddress 0030h).

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9 Limiting values

Table 12 describes the limiting values for SC18IS606.

Table 12. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134).[1][2]

Symbol	Parameter	Conditions	Note	Min	Max	Unit
V_{DD}	Supply voltage			-0.5	+4.6	V
$T_{amb(bias)}$	Bias ambient temperature	Operating		-55	+150	°C
T_{stg}	Storage temperature			-65	+150	°C
V_I	Input voltage	5 V tolerant I/O pins; $V_{DD} \geq 1.71$ V	[3] [4]	-0.5	+5.4	V
		3 V tolerant I/O - all pins	[5]	-0.5	+3.6	V
$I_{OH(I/O)}$	HIGH-level output current per input/output pin			-	8	mA
$I_{OL(I/O)}$	LOW-level output current per input/output pin			-	20	mA
$I_{I/O(tot)(max)}$	Maximum total I/O current			-	120	mA
P_{tot}/pack	Total power dissipation per package		[6]	-	1.5	W

[1] This product includes circuitry designed for the protection of its internal devices from the damaging effects of excessive static charge. Nonetheless, it is suggested that conventional precautions be taken to avoid applying greater than the rated maximum.
[2] Parameters are valid over the operating temperature range unless otherwise specified. All voltages are with respect to V_{SS} unless otherwise noted.
[3] Applies to all 5 V tolerant I/O pins except the 3 V tolerant pin CITO.
[4] Including the voltage on outputs in 3-state mode.
[5] V_{DD} is present or not present.
[6] Based on package heat transfer, not device power consumption.

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10 Static characteristics

Table 13 describes the static characteristics for SC18IS606.

Table 13. Static characteristics

V_{DD} = 1.71 V to 3.6 V; T_{amb} = -40 °C to +105 °C, unless otherwise specified

Symbol	Parameter	Conditions	Note	Min	Typ [1]	Max	Unit
V_{DD}	Supply voltage			1.71		3.6	V
$I_{DD(oper)}$	Operating supply current	V_{DD} = 3.3 V		-	2.1	4.0	mA
$I_{DD(idle)}$	Idle mode supply current	V_{DD} = 3.3 V		-	1.3	3.4	mA
$V_{th(HL)}$	HIGH-LOW threshold voltage	Schmitt trigger input		0.22V_{DD}	0.4V_{DD}	-	V
$V_{th(LH)}$	LOW-HIGH threshold voltage	Schmitt trigger input		-	0.6V_{DD}	0.7V_{DD}	V
V_{hys}	Hysteresis voltage			-	0.2V_{DD}	-	V
V_{OL}	LOW-level output voltage	All pins					
		I_{OL} = 4 mA; 2.5 V ≤ V_{DD} ≤ 3.6 V		-		0.5	V
		I_{OL} = 3 mA; 1.71 V ≤ V_{DD} ≤ 2.5 V		-		0.5	V
V_{OH}	HIGH-level output voltage	All pins					
		I_{OH} = 4 mA; 2.5 V ≤ V_{DD} ≤ 3.6 V		V_{DD} - 0.4	-	-	V
		I_{OH} = 3 mA; 1.71 V ≤ V_{DD} ≤ 2.5 V		V_{DD} - 0.5		-	V
C_{ig}	Input capacitance at the gate		[2]	-	-	15	pF
I_{IL}	LOW-level input current	Logical 0; V_I = 0.4 V		-	-	-80	µA
I_{LI}	Input leakage current	All ports; V_I = V_{IL} or V_{IH}	[3]	-	-	±10	µA
I_{THL}	HIGH-LOW transition current	All ports; logical 1-to-0; V_I = 2.0 V at V_{DD} = 3.6 V		-30	-	-450	µA
I_{pu}	Pull-up current	V_I = 0 V;	[4]				µA
		2.0 V ≤ V_{DD} ≤ 3.6 V		10	50	90	µA
		1.71 V ≤ V_{DD} ≤ 2.0 V		7	50	85	µA
		V_{DD} < V_I < 5 V		0	0	0	µA

[1] Typical ratings are not guaranteed. The values listed are at room temperature, 3 V.
[2] Pin capacitance is characterized but not tested.
[3] Measured with pins in high-impedance mode.
[4] Pullup current measured across the weak pullup resistor

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11 Dynamic characteristics

Table 14 describes the dynamic characteristics for SC18IS606.

Table 14. Dynamic characteristics

V_{DD} = 1.71 V to 3.6 V; T_{amb} = -40 °C to +105 °C, unless otherwise specified

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Glitch filter						
t_{gr}	Glitch rejection time	$\overline{\text{RESET}}$ pin	-	-	20	ns
t_{sa}	Signal acceptance time	$\overline{\text{RESET}}$ pin	125	-		ns
		Any pin except $\overline{\text{RESET}}$	50	-	-	ns
SPI controller interface						
f_{SPI}	SPI operating frequency	1.875 MHz	-	-	1.875	MHz
T_{SPICYC}	SPI cycle time	1.875 MHz	533	-	-	ns
t_{DS}	Data set-up time		10	-	-	ns
t_{DH}	Data hold time		7	-	-	ns
$t_{V(Q)}$	Data output valid time		-	-	2	ns

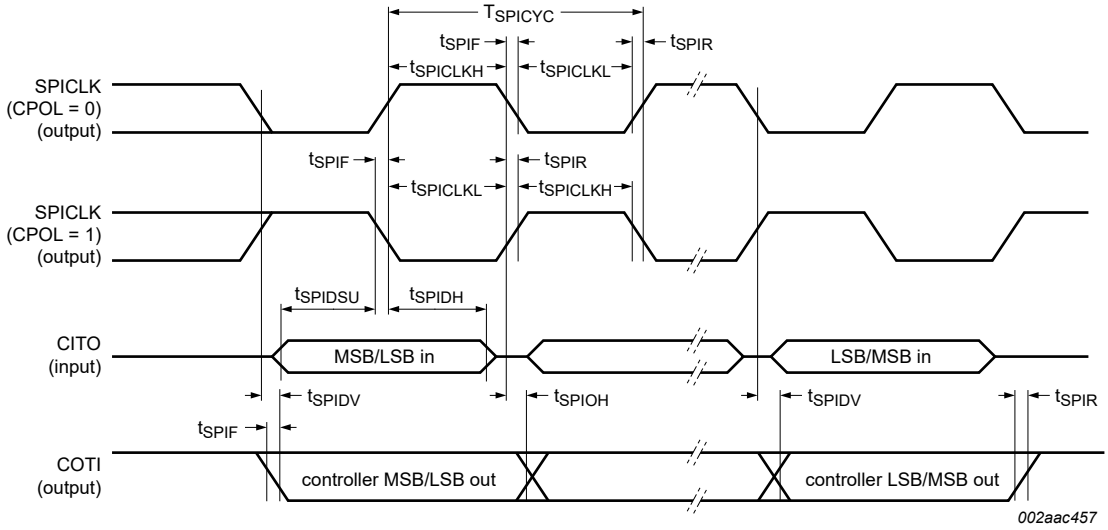


Figure 18. SPI controller timing (CPHA = 0)

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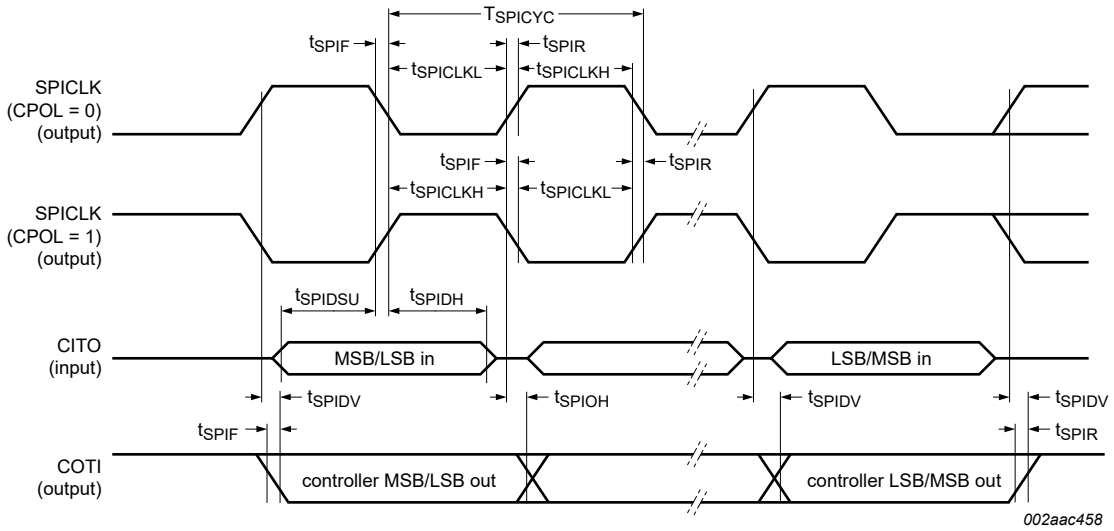


Figure 19. SPI controller timing (CPHA = 1)

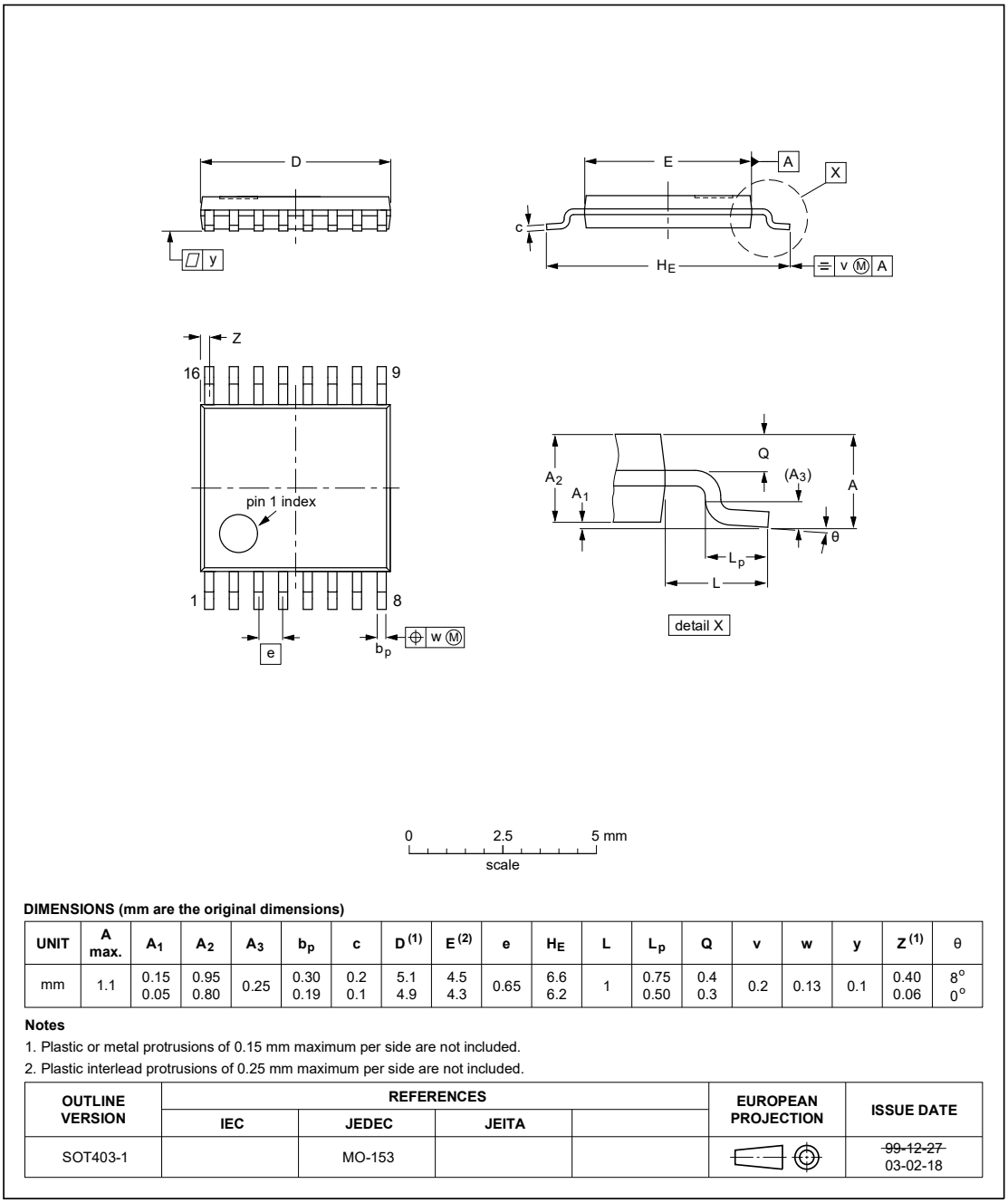
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12 Package outline

Figure 20 shows the package outline for the SC18IS606.

TSSOP16: plastic thin shrink small outline package; 16 leads; body width 4.4 mm **SOT403-1**



DIMENSIONS (mm are the original dimensions)

UNIT	A max.	A_1	A_2	A_3	b_p	c	D (1)	E (2)	e	H_E	L	L_p	Q	v	w	y	Z (1)	θ
mm	1.1	0.15 0.05	0.95 0.80	0.25	0.30 0.19	0.2 0.1	5.1 4.9	4.5 4.3	0.65	6.6 6.2	1	0.75 0.50	0.4 0.3	0.2	0.13	0.1	0.40 0.06	8° 0°

Notes

1. Plastic or metal protrusions of 0.15 mm maximum per side are not included.
2. Plastic interlead protrusions of 0.25 mm maximum per side are not included.

OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	JEITA			
SOT403-1		MO-153				~~99-12-27~~ 03-02-18

Figure 20. Package outline SOT403-1 (TSSOP16)

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13 Printed Circuit Board (PCB) layout

This section shows the PCB layout for the SC18IS606.

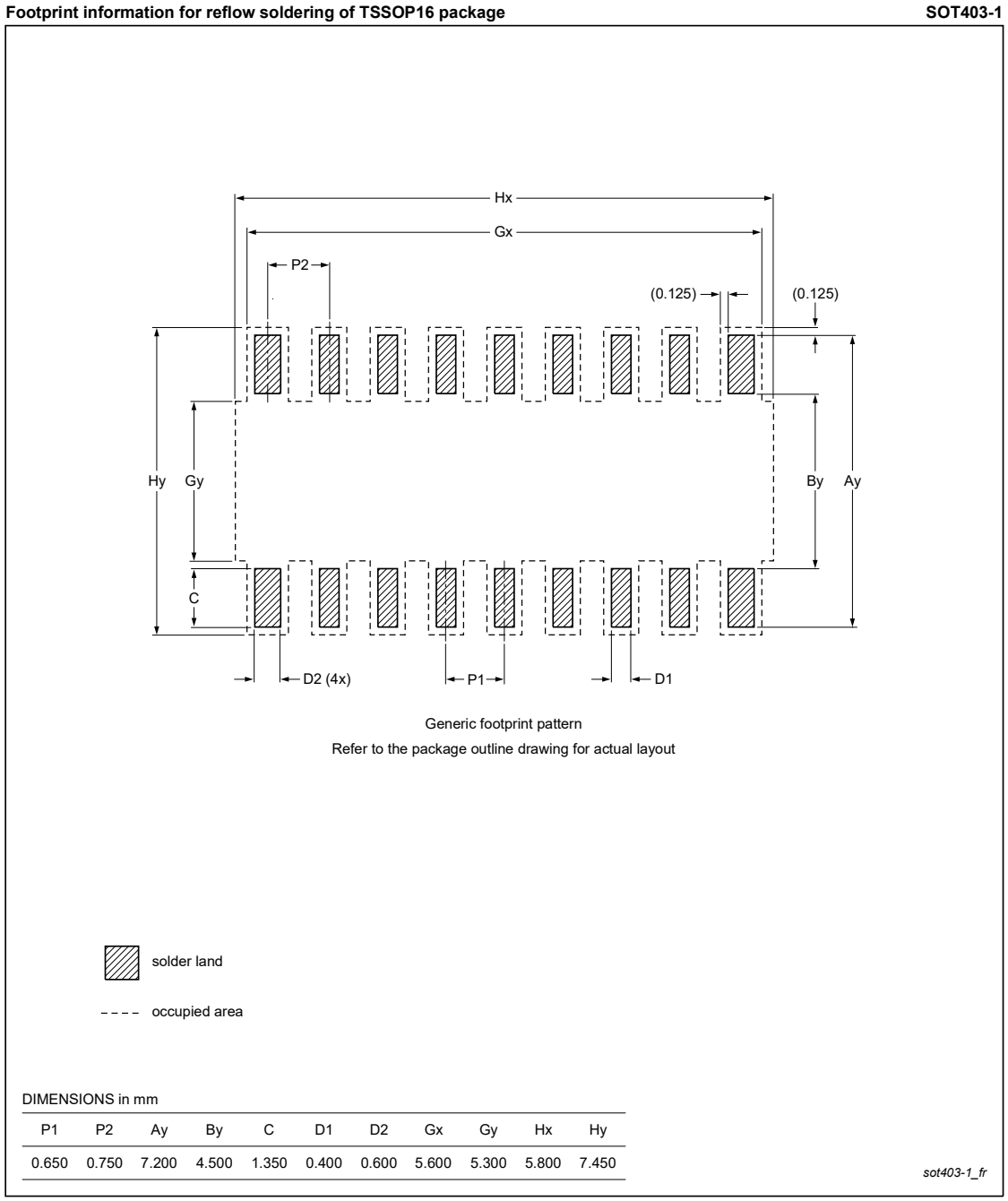


Figure 21. SOT403-1 (TSSOP16) footprint information for reflow soldering

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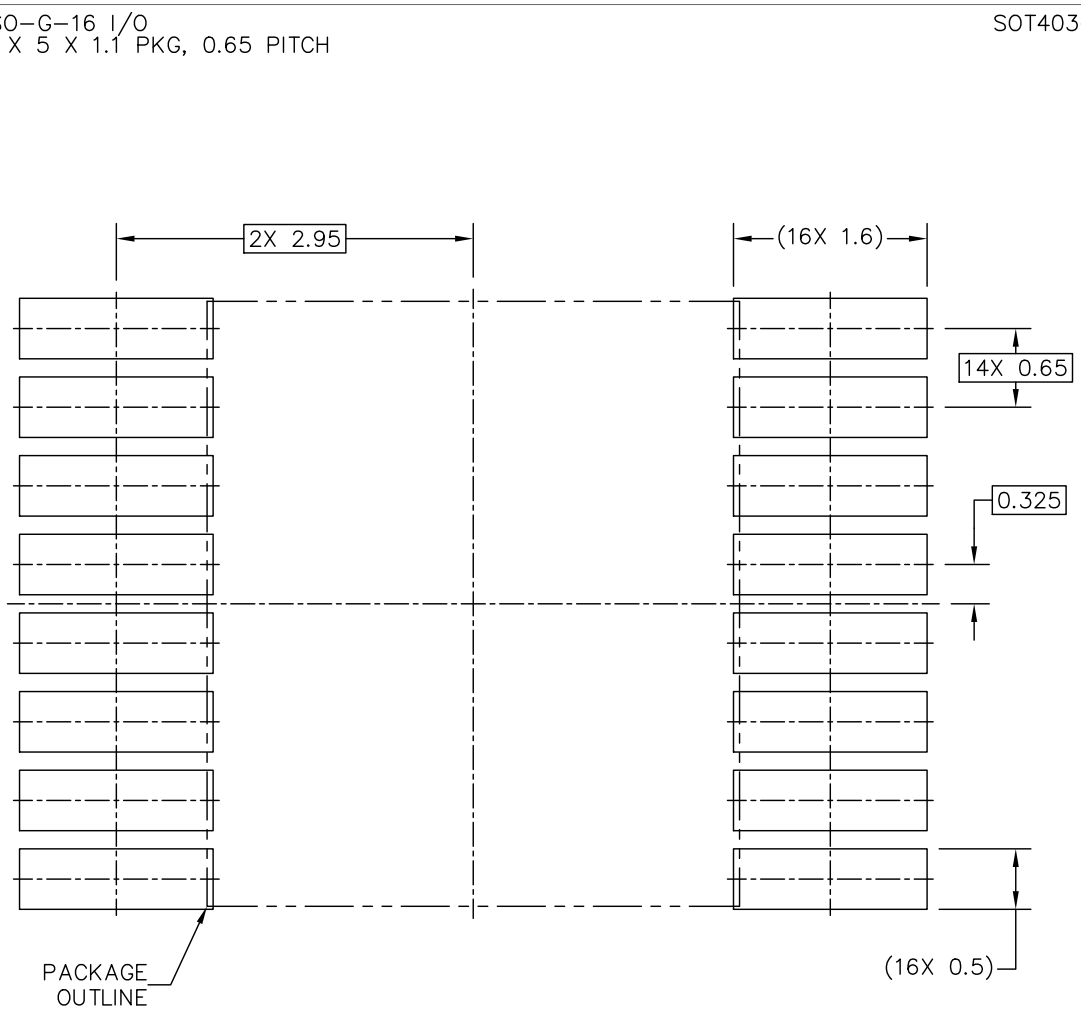


Figure 22. SOT403-1 (TSSOP16) solder mask opening pattern

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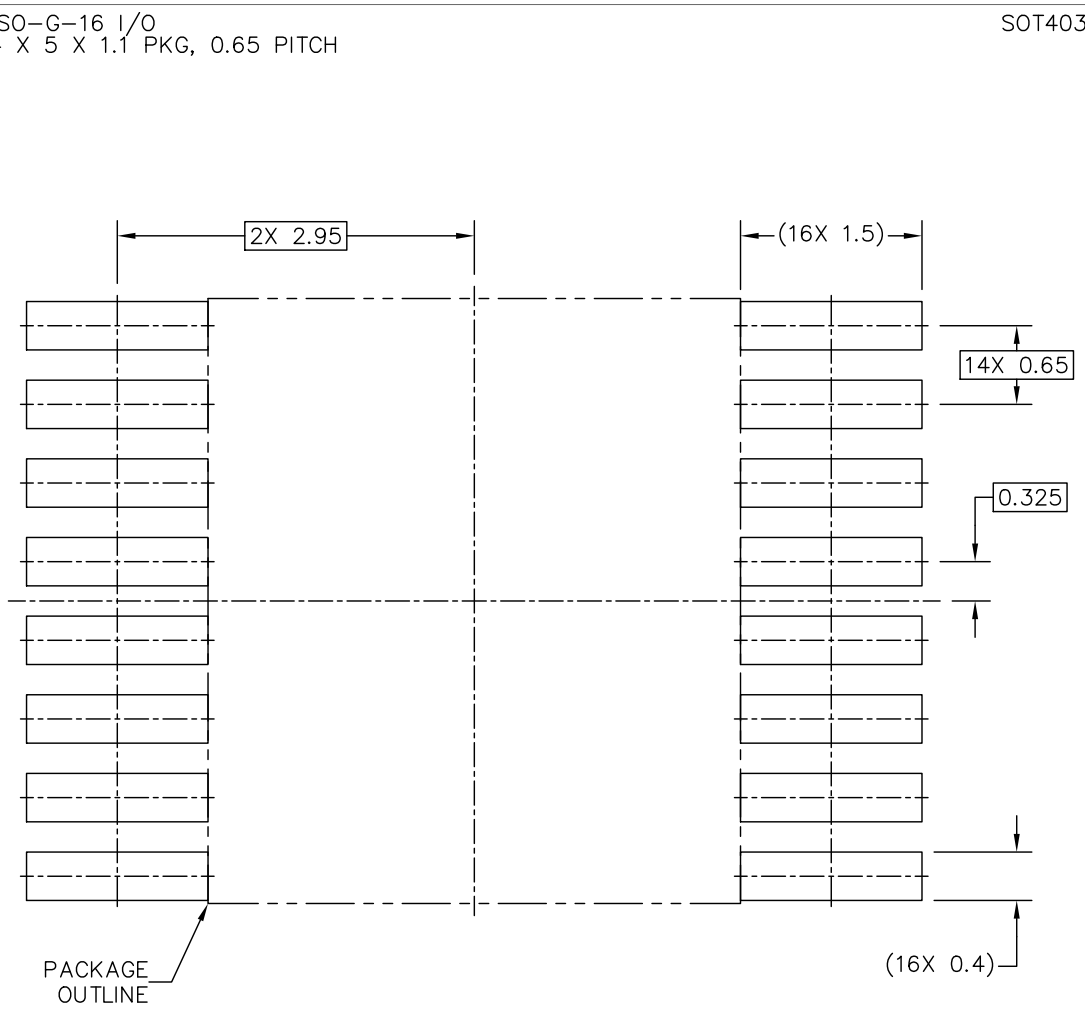


Figure 23. SOT403-1 (TSSOP16) I/O pads and solderable area

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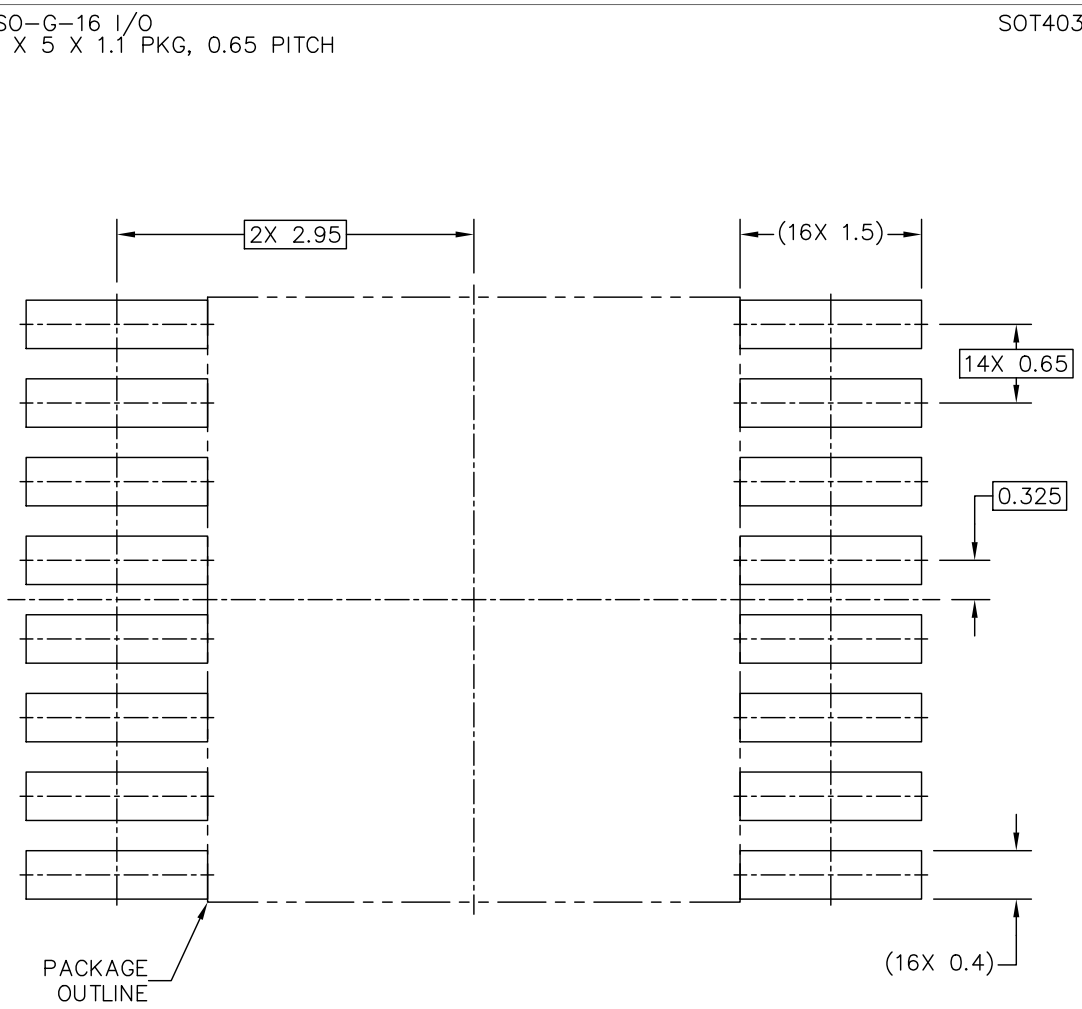


Figure 24. SOT403-1 (TSSOP16) Solder paste stencil

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14 Soldering of SMD packages

This text provides a very brief insight into a complex technology. A more in-depth account of soldering ICs can be found in Application Note *AN10365 "Surface mount reflow soldering description"*.

14.1 Introduction to soldering

Soldering is one of the most common methods through which packages are attached to Printed Circuit Boards (PCBs), to form electrical circuits. The soldered joint provides both the mechanical and the electrical connection. There is no single soldering method that is ideal for all IC packages. Wave soldering is often preferred when through-hole and Surface Mount Devices (SMDs) are mixed on one printed wiring board; however, it is not suitable for fine pitch SMDs. Reflow soldering is ideal for the small pitches and high densities that come with increased miniaturization.

14.2 Wave and reflow soldering

Wave soldering is a joining technology in which the joints are made by solder coming from a standing wave of liquid solder. The wave soldering process is suitable for the following:

- Through-hole components
- Leaded or leadless SMDs, which are glued to the surface of the printed circuit board

Not all SMDs can be wave soldered. Packages with solder balls, and some leadless packages which have solder lands underneath the body, cannot be wave soldered. Also, leaded SMDs with leads having a pitch smaller than ~0.6 mm cannot be wave soldered, due to an increased probability of bridging.

The reflow soldering process involves applying solder paste to a board, followed by component placement and exposure to a temperature profile. Leaded packages, packages with solder balls, and leadless packages are all reflow solderable.

Key characteristics in both wave and reflow soldering are:

- Board specifications, including the board finish, solder masks and vias
- Package footprints, including solder thieves and orientation
- The moisture sensitivity level of the packages
- Package placement
- Inspection and repair
- Lead-free soldering versus SnPb soldering

14.3 Wave soldering

Key characteristics in wave soldering are:

- Process issues, such as application of adhesive and flux, clinching of leads, board transport, the solder wave parameters, and the time during which components are exposed to the wave
- Solder bath specifications, including temperature and impurities

14.4 Reflow soldering

Key characteristics in reflow soldering are:

- Lead-free versus SnPb soldering; note that a lead-free reflow process usually leads to higher minimum peak temperatures (see Figure 25) than a SnPb process, thus reducing the process window
- Solder paste printing issues including smearing, release, and adjusting the process window for a mix of large and small components on one board

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- Reflow temperature profile; this profile includes preheat, reflow (in which the board is heated to the peak temperature) and cooling down. It is imperative that the peak temperature is high enough for the solder to make reliable solder joints (a solder paste characteristic). In addition, the peak temperature must be low enough that the packages and/or boards are not damaged. The peak temperature of the package depends on package thickness and volume and is classified in accordance with Table 15 and Table 16

Table 15. SnPb eutectic process (from J-STD-020D)

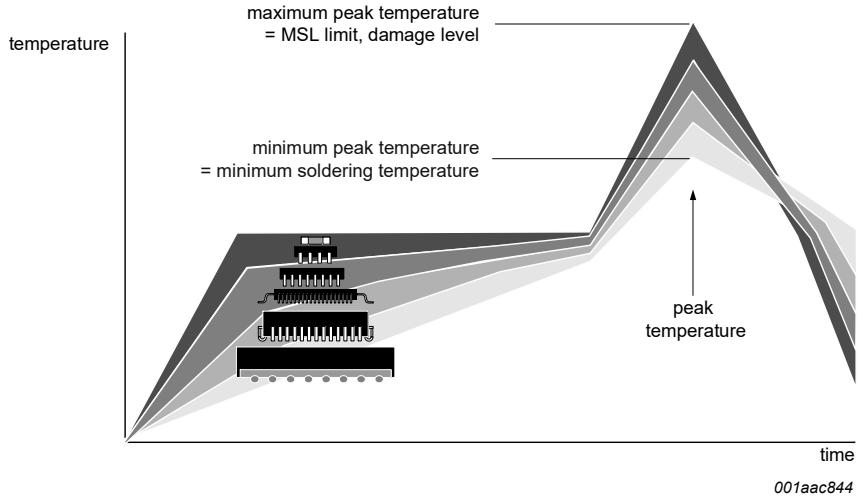
Package thickness (mm)	Package reflow temperature (°C)	
	Volume (mm³)	
	< 350	≥ 350
< 2.5	235	220
≥ 2.5	220	220

Table 16. Lead-free process (from J-STD-020D)

Package thickness (mm)	Package reflow temperature (°C)		
	Volume (mm³)		
	< 350	350 to 2000	> 2000
< 1.6	260	260	260
1.6 to 2.5	260	250	245
> 2.5	250	245	245

Moisture sensitivity precautions, as indicated on the packing, must be respected at all times.

Studies have shown that small packages reach higher temperatures during reflow soldering, see Figure 25.



MSL: Moisture Sensitivity Level

Figure 25. Temperature profiles for large and small components

For further information on temperature profiles, refer to Application Note *AN10365 "Surface mount reflow soldering description"*.

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15 Acronyms

Table 17 lists the acronyms used in this document.

Table 17. Acronyms

Acronym	Description
COTI	Controller Output Target Input
CPU	Central Processing Unit
CITO	Controller Input Target Output
CS	Chip Select
EEPROM	Electrically Erasable Programmable Read-Only Memory
ESD	Electrostatic Discharge
GPIO	General-Purpose Input/Output
HBM	Human Body Model
I/O	Input/Output
I^2C-bus	Inter-Integrated Circuit bus
LSB	Least Significant Bit
MSB	Most Significant Bit
PCB	Printed Circuit Board
SCL	Serial Clock
SDA	Serial Data
SMD	Surface Mounted Device
SPI	Serial Peripheral Interface

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16 Revision history

Table 18 summarizes revisions to this document.

Table 18. Revision history

Document ID	Release date	Description
SC18IS606 v.1.2	13 October 2026	Updated as per CIN# 202605002I: • Figure 1 "Block diagram of SC18IS606" • Figure 2 "Pin configuration for TSSOP16"
SC18IS606 v.1.1	27 May 2026	Updated as per CIN# 202605002I: • Removed IDD(tpd) parameter related information in Section 10 "Static characteristics". • Minor editorial changes
SC18IS606 v.1.0	15 September 2021	Initial public release

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Document status[1][2]	Product status[3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
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Product [short] data sheet	Production	This document contains the product specification.

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